

birdclef-2026

11th Place Solution (and the 3rd Place that Got Away) [Without Perch]

Rank	11
Team	Oh Captain! My Captain!
Author	Salman Ahmed
Topic ID	704264
Version	Original

Source: <https://www.kaggle.com/competitions/birdclef-2026/discussion/704264>
Retrieved with Kaggle CLI on 2026-07-03.

Concern: I think this year, there was too much noise not just in Kaggle Notebooks but in Discussions. I was unable to share anything with the Kaggle Competition community and unable to make my solutions public. I planned to make my solution public a month ago, but each day I saw tens of new notebooks and some random discussion posts. ☹

Let me share my solution. I'll keep it simple.

My personal GPU rig got fried a couple of weeks before the competition, and I only had access to a single small personal GPU at home. I didn't want to go through the hassle of setting the rig up again. Google's Perchv2 got a lot of attention, and I initially thought to finetune that at the start of the competition, but I couldn't fit it in my GPU. Later, I borrowed a GPU from a friend, but by then I couldn't plan to finetune Perch.

I know a lot of people train models with small Mel spectrograms because they contain enough information, but I think small models like EfficientNet-B0 are not as perfect as they should be considering their receptive field. To help, I like the idea of increasing Mels. Even though it increases compute, I couldn't afford to pre-train a better architecture with small Mels. @coolz did the same in previous competitions, so I followed that approach. Ultimately, I saw that increasing Mels beyond 480 didn't help, so I stopped there.

Melspec Settings

- Mels: 480
- Hop Length: 1001
- N_FFT: 4096

The reason for the 1001 hop length is that for 20 seconds of audio, I wanted to get exactly 20 frame probabilities as outputs, basically a single framewise probability for each second.

Code Repository <https://github.com/sahmedcodes/birdclef26-weekend-11thplace-3rd>

Inference Notebook <https://www.kaggle.com/code/salmanahmedtam/pow-fork-of-frame-ens-stage3-eff-family-20sec>

Training and Inference Strategy

- All models are trained with CrossEntropyLoss, and Sigmoid is used for inference.
- All models are trained with 'nocall' as an additional class.

Model Stages and Validation Performance

- **Stage 0 Models** (Train Audios Only)
 - **Architectures:** B0_NS + V2B0 + V1B1 + B1NS
 - **Public Score:** 0.936
 - **Private Score:** 0.944
- **Stage 1 Models** (Train Audios Only)
 - **Architectures:** B0_NS + V2B0 + V1B1
 - **Public Score:** 0.943
 - **Private Score:** 0.959

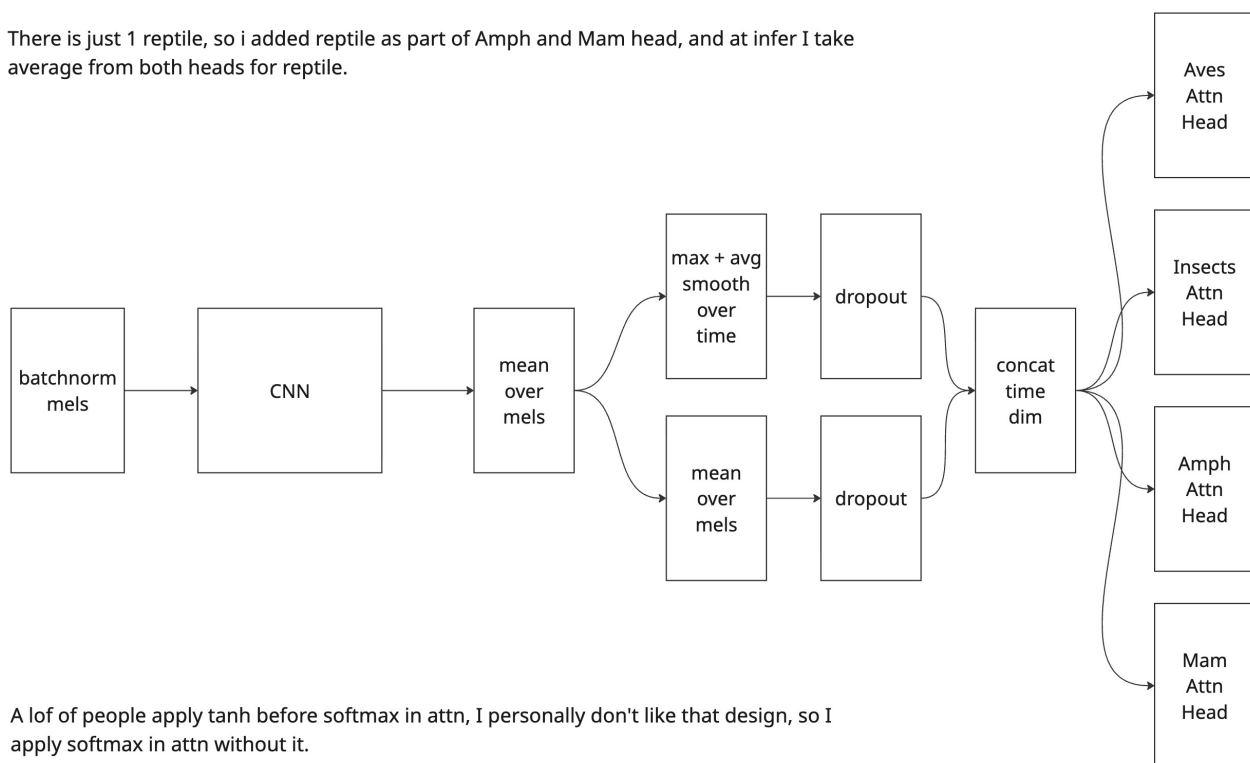
- **Note:** This missed submission would have achieved 3rd place.
- **Stage 2 Models** (Train Audios + Train Soundscapes)
- **Architectures:** B0_NS + B1_NS + LITE1 + LITE2
- **Public Score:** 0.948
- **Private Score:** 0.958
- **Note:** Missed submission solutions.
- **Stage 3 Models** (Train Audios + Train Soundscapes + XC Additional Data)
- **Architectures:** B0_NS

Selected Final Submission Ensemble

- Composition: STAGE0_B0 + STAGE1_B0 + STAGE0_B1 + STAGE2_B0 + STAGE2_B2 + STAGE3_B0
- Public Score: 0.953 | Private Score: 0.955

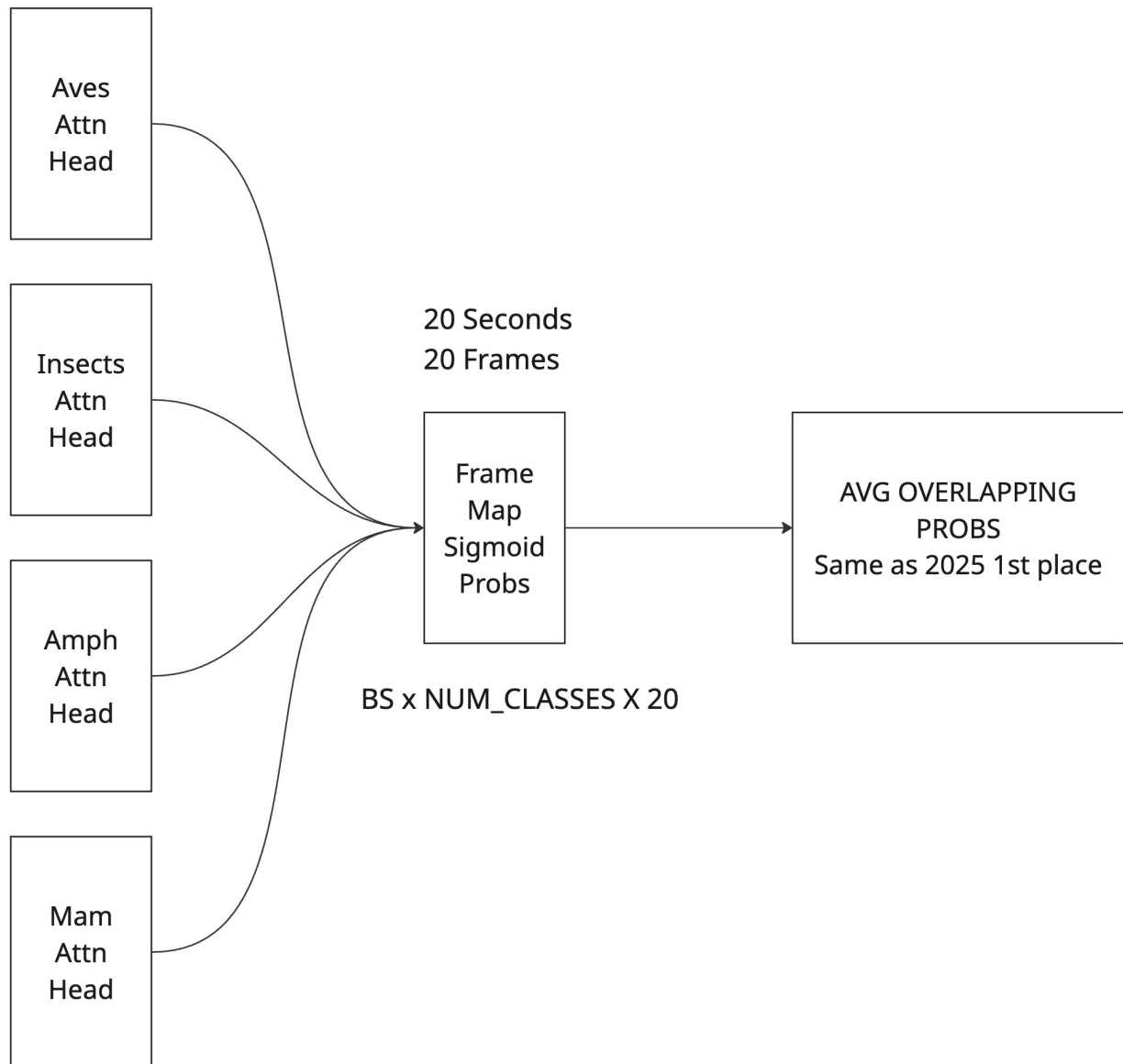
Model Architecture

There is just 1 reptile, so i added reptile as part of Amph and Mam head, and at infer I take average from both heads for reptile.



A lot of people apply tanh before softmax in attn, I personally don't like that design, so I apply softmax in attn without it.

Inference Single Model Pipeline



Missed Submissions Leaderboard Position

✓	<u>Post stage1 3 models ensemble - Version 2</u> Succeeded · 1mo ago	0.959	0.943	☐
✓	<u>Fork of stage3x_ens_20sec_batchwise_b0b1lite1lite2 - Version 3</u> Succeeded · 3d ago	0.958	0.948	☐
✓	<u>frame_ens_stage3_eff_family_20sec_xc - Version 1</u> Succeeded · 4d ago	0.958	0.949	☐
✓	<u>stage1 3 models ensemble - Version 1</u> Succeeded · 1mo ago	0.957	0.944	☐

I probably missed a lot of stuff. Ping me if you have any questions.

Last year, I didn't document the details, so I forgot what I did and why I did it. I am adding it here so I remember next time.

Supplemental Q&A

Kurise · 2026-06-04T03:17:34.003000

Congratulations on your solo gold medal! Regarding the loss function, may I ask if you tried different losses like last year? Last year, switching losses made a big performance difference, but this year when I tried several, the results were nearly the same. Is that why your model trained only using CrossEntropyLoss?

Salman Ahmed · 2026-06-04T03:21:53.950000

I tried FocalBCE / BCE, but based on the architecture I planned, with an additional no species class for each head CrossEntropy made more sense.

For example; if the example had just Aves in it, then no-call class would be 1.0 in insects / mam / amph head. and CrossEntropy would be better for that case, compared to BCE.

```
insects_targets[insects_targets[:, :-1].sum(dim=1) < 0.5, -1] = 1
```

chenwenqiang_001 · 2026-06-04T03:45:43.470000

Could you show more details about training, augumention,mixup?

Salman Ahmed · 2026-06-04T03:47:30.833000

It's in the github repo shared in the solution write up. I just used mixup augmentation, nothing else.

yanqiangmiffy · 2026-06-04T04:06:54.217000

Congratulations ! How to obtain XC additional data? Could you share the collection process?

Salman Ahmed · 2026-06-04T04:09:42.623000

It's in the github repo I shared. FYI, that additional data didn't help at all. Infact it had a reverse effect. But I kept the model anyway for diversity.